

# The Markowitz Portfolio



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# Managerial motivation

# Introduction: Managerial motivation

- ▶ How much should we invest in each asset in order to balance **expected return** and **risk**?
- ▶ The Markowitz model is the **most famous QP in history**: it minimizes the variance of the portfolio subject to a minimum return
- ▶ Class of the problem: **convex QP**

# The efficient frontier

- ▶ The result is not a single number but an **efficient frontier**
- ▶ It is the complete menu of risk-return trade-offs
- ▶ The decision maker chooses on this frontier according to their own degree of **risk aversion**

# A structure that reappears everywhere

- ▶ The problem was born in finance (Markowitz, 1952, Nobel prize)
- ▶ Its structure — **allocating a budget among risky and correlated alternatives** — reappears everywhere:
  - ▶ mixes of R&D projects
  - ▶ customer portfolios
  - ▶ allocation of production capacity across volatile markets
- ▶ It is the natural bridge between optimization and statistics: the data of the model are **estimates**, with all the consequences that this entails

# In this chapter we will learn how to

1. formulate and solve a **convex QP**
2. build and read the **efficient frontier**
3. understand the effect of the **mandate constraints**
4. **be wary** of the estimates of the expected returns

# Summary

- ▶ Markowitz: minimize the risk (variance) while guaranteeing a minimum return
- ▶ The result is an **efficient frontier** of risk-return trade-offs
- ▶ The data are **estimates**: optimization meets statistics

# Guided construction of the model



# Introduction: Guided construction of the model

- ▶ We describe the problem in words, then we fix **data** and **variables**
- ▶ We write the **minimum-variance QP model** and explain objective and constraints
- ▶ We see two **equivalent formulations** and why the QP is **convex**

# The problem in words

- ▶ **We decide** the share of capital to invest in each asset
- ▶ **The objective** is to minimize the risk of the portfolio, measured by the variance of its return
- ▶ **The constraints:**
  - ▶ invest the whole capital (the shares sum to 1)
  - ▶ guarantee a minimum expected return
  - ▶ respect the minimum and maximum share of each asset

## Data (input of the model) (1/2)

| Symbol   | Type                      | Meaning                                                                                                                                       |
|----------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $n$      | $\in \mathbb{Z}_{\geq 1}$ | number of assets; the assets are indexed by $i \in \{1, 2, \dots, n\}$                                                                        |
| $\mu_i$  | $\in \mathbb{Q}$          | expected return of asset $i$ (per year)                                                                                                       |
| $q_{ij}$ | $\in \mathbb{Q}$          | covariance between the returns of assets $i$ and $j$ ; the matrix $\mathbf{Q} = (q_{ij})$ is positive semidefinite ( $\mathbf{Q} \succeq 0$ ) |

## Data (input of the model) (2/2)

| Symbol        | Type                                            | Meaning                                                     |
|---------------|-------------------------------------------------|-------------------------------------------------------------|
| $\bar{r}$     | $\in \mathbb{Q}$                                | minimum required expected return                            |
| $\ell_i, u_i$ | $\in \mathbb{Q}, 0 \leq \ell_i \leq u_i \leq 1$ | minimum and maximum share that can be invested in asset $i$ |

# Decision variables

- ▶ We introduce the following  $n$  non-negative variables:

$x_i$  = share of capital invested in asset  $i$ ,  $\forall i \in \{1, 2, \dots, n\}$

- ▶ Using these variables, we write a QP model for the problem

## Definition 1: Minimum-variance QP model with a minimum return

$$\min \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j$$

$$\text{s.t.} \quad \sum_{i=1}^n \mu_i x_i \geq \bar{r}$$

$$\sum_{i=1}^n x_i = 1$$

$$x_i \leq u_i, \quad \forall i \in \{1, \dots, n\}$$

$$x_i \geq \ell_i, \quad \forall i \in \{1, \dots, n\}$$

# The objective function

- ▶ The **quadratic** objective function minimizes the variance of the return of the portfolio
- ▶ It contains the covariances  $q_{ij}$ , and this is where **diversification** comes from
- ▶ Combining weakly correlated assets lowers the overall risk **below that of the individual components**

# Minimum return and budget

- ▶ The linear constraint  $\sum_i \mu_i x_i \geq \bar{r}$  imposes that the expected return of the portfolio reaches the threshold  $\bar{r}$  (one linear constraint)
  - ▶ it is the “**dial**” with which the efficient frontier is traced out
  - ▶ its multiplier tells us how much variance each additional point of return costs
- ▶ The linear constraint  $\sum_i x_i = 1$  imposes that the whole capital is invested (one linear constraint)

## Bounds per asset

- ▶ The linear constraints  $x_i \leq u_i$  are the **caps per asset**, typical regulatory or mandate constraints ( $n$  linear constraints)
- ▶ The constraints  $x_i \geq \ell_i$  define the variables of the model
- ▶ With  $\ell_i = 0$ : **short selling is forbidden**

# Equivalent formulations

**Mean-variance:** the risk goes into the objective with a weight  $\lambda$

$$\max \sum_{i=1}^n \mu_i x_i - \lambda \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j$$

**Maximum return with bounded risk:** the risk goes into the constraints

$$\max \sum_{i=1}^n \mu_i x_i \quad \text{with variance} \leq \bar{\sigma}^2$$

By varying  $\bar{r}$  (or  $\lambda$ , or  $\bar{\sigma}$ ) **the same frontier** is traced out.

# Convexity

- ▶ Every covariance matrix is **positive semidefinite**
- ▶ Hence the QP is **convex**
- ▶ The optimum that is found is a **certified global** one

# Summary

- ▶ Variables  $x_i$ : shares of capital invested in the assets
- ▶ Quadratic objective (variance); linear constraints: minimum return, budget, bounds per asset
- ▶ Three equivalent formulations (minimum variance, mean-variance, bounded risk) generate the same frontier
- ▶  $Q \succeq 0$ : the QP is convex, the optimum is global

# Worked example by hand

# Introduction: Worked example by hand

- ▶ Only two assets, uncorrelated
- ▶ We compute by hand the **minimum-variance** portfolio
- ▶ We will see the **diversification effect** in its purest form

## Example: Two uncorrelated assets (1/2)

Volatilities  $\sigma_1 = 20\%$ ,  $\sigma_2 = 30\%$ , zero correlation.

Minimize the variance with  $x_1 + x_2 = 1$  (without the return constraint).

**Step 1.** Substituting  $x_2 = 1 - x_1$ :

$$f(x_1) = 0.04 x_1^2 + 0.09 (1 - x_1)^2$$

**Step 2.**

$$f'(x_1) = 0.08 x_1 - 0.18 (1 - x_1) = 0$$

$$\Rightarrow x_1 = \frac{0.09}{0.04 + 0.09} = \frac{9}{13} \approx 69.2\%$$

## Example: Two uncorrelated assets (2/2)

**Step 3.** Variance of the portfolio:

$$f(9/13) = 0.04 \cdot 0.479 + 0.09 \cdot 0.095 = 0.0277$$

that is a volatility of 16.6%

- ▶ The portfolio is **less risky than either of the two assets**
- ▶ This is the diversification effect in its purest form: the optimal weights are **inversely proportional to the variances**

# Summary

- ▶ With two uncorrelated assets the minimum is found by setting the derivative to zero
- ▶  $x_1 = 9/13$ : volatility 16.6%, lower than that of either asset
- ▶ Optimal weights inversely proportional to the variances

# Case study

# Introduction: Case study

- ▶ Eight sector ETFs with simulated monthly returns
- ▶  $\mu$  and  $Q$  are **estimated** from the data, as in practice
- ▶ We see the model written in **Python with Gurobi**

# The data of the case study

- ▶ Eight **sector ETFs** (*Exchange Traded Fund*: listed funds that replicate a basket of securities of a given sector)
- ▶ 60 monthly returns simulated with a single-market-factor model
- ▶ File `markowitz_rendimenti.csv`
- ▶  $\mu$  and  $Q$  are **estimated** from these data, as in practice

# Annualized statistics

| Asset | $\mu$  | vol    | Asset | $\mu$  | vol    |
|-------|--------|--------|-------|--------|--------|
| ENE   | 19.12% | 18.65% | SAN   | 2.80%  | 11.93% |
| FIN   | 8.27%  | 19.59% | CON   | 11.25% | 11.67% |
| TEC   | 1.58%  | 29.27% | UTL   | 11.99% | 7.53%  |
| IND   | 9.97%  | 16.60% | MAT   | 20.93% | 23.93% |

# The model in Gurobi

```
m = gp.Model("markowitz")
# shares,  $0 \leq x_i \leq u$ 
x = m.addVars(n, ub=u, name="x")
m.addConstr(x.sum() == 1, name="budget")
m.addConstr(gp.quicksum(mu[i]*x[i]
                        for i in range(n)) >= r_min)
m.setObjective(gp.quicksum(Q[i, j]*x[i]*x[j]
                        for i in range(n)
                        for j in range(n)),
               GRB.MINIMIZE)
m.optimize()
```

# Summary

- ▶ Eight sector ETFs, 60 months of simulated returns
- ▶ Expected returns and covariances are **estimates** from the data
- ▶ In Gurobi the quadratic objective is written directly with quicksum over the products  $x_i x_j$

# Results

# Introduction: Results

- ▶ We compare the **minimum-variance** portfolio with the **equally weighted**  $1/n$  one
- ▶ We draw the **efficient frontier**, with and without caps per asset
- ▶ We look at how the optimal **composition** changes along the frontier

## Minimum variance versus equally weighted

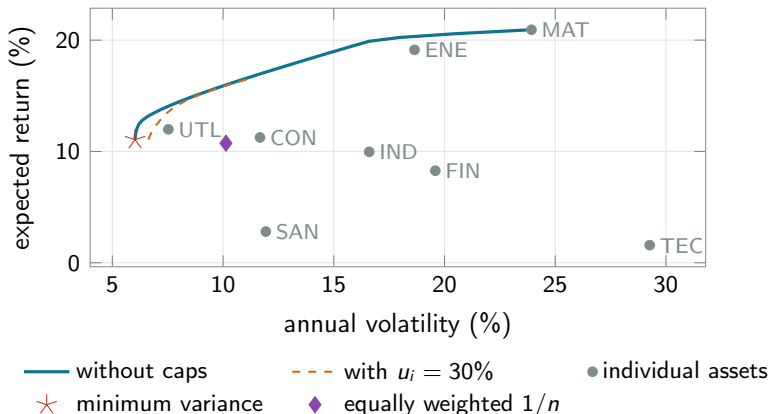
| Portfolio                  | Return | Volatility |
|----------------------------|--------|------------|
| Global minimum variance    | 11.06% | 6.03%      |
| Equally weighted ( $1/n$ ) | 10.74% | 10.13%     |

### Minimum-variance composition:

| ENE  | IND  | SAN   | CON   | UTL   |
|------|------|-------|-------|-------|
| 5.2% | 1.9% | 11.5% | 22.3% | 58.3% |

# Efficient frontier

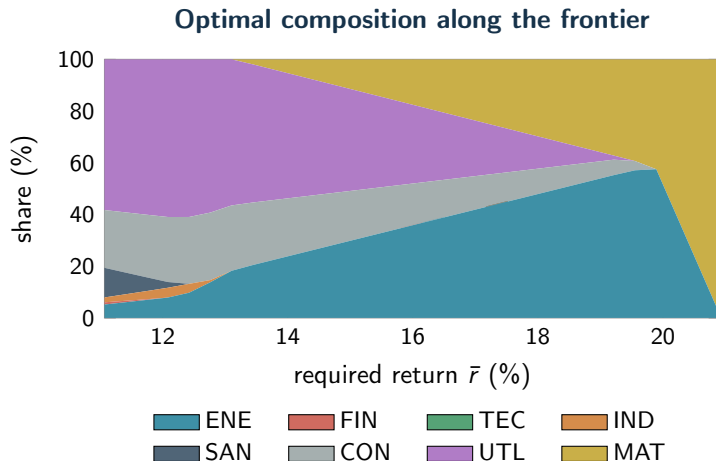
## Efficient frontier: diversification dominates the individual assets



# How to read the risk-return plane

- ▶ The eight assets: **grey dots**
- ▶ Efficient frontier without caps: **teal line**
- ▶ Frontier with a 30% cap per asset: **dashed orange**
- ▶ Minimum-variance portfolio: **red star**
- ▶ Equally weighted  $1/n$ : **purple diamond**, which turns out to be **dominated**

# Optimal composition along the frontier



# How to read the composition

- ▶ Stacked areas (in %) as the required return  $\bar{r}$  grows
- ▶ On the left the **defensive mix** dominates (UTL, CON, SAN)
- ▶ Towards the right the portfolio concentrates progressively on the **high-return** assets (MAT, ENE)
- ▶ At the extreme: the **single-asset** portfolio

## Three messages from the figure

1. The minimum-variance portfolio (volatility 6%) is much less risky than the best individual asset (UTL, 7.5%) while still returning 11%
2. The equally weighted  $1/n$  portfolio — the “I know nothing” strategy — is clearly **dominated**: the same region of return but almost twice the volatility
3. The cap  $u_i = 30\%$  cuts off the upper part of the frontier: the **cost of the mandate constraints** can be seen as the distance between the two curves

# Summary

- ▶ Minimum variance: return 11.06%, volatility 6.03%
- ▶ The equally weighted portfolio is dominated: similar return, almost twice the volatility
- ▶ Along the frontier we move from the defensive mix to the high-return assets
- ▶ The mandate constraints have a visible cost on the frontier

# Sensitivity analysis

# Introduction: Sensitivity analysis

- ▶ We let the minimum required return  $\bar{r}$  vary
- ▶ We will understand when the return constraint is **active** and how much it costs
- ▶ Finally we will see the **fragility of the estimates** of the expected returns

## Volatility as $\bar{r}$ varies

| $\bar{r}$ | Volatility |                                                    |
|-----------|------------|----------------------------------------------------|
| 6%        | 6.03%      | constraint <b>not</b> active: same as min variance |
| 8%        | 6.03%      | same                                               |
| 10%       | 6.03%      | same                                               |
| 12%       | 6.10%      | $d(\text{variance})/d(\bar{r}) \approx 0.0221$     |

# A constraint that costs nothing

- ▶ Up to  $\bar{r} = 11\%$  the return constraint is **inactive**
- ▶ The minimum-variance portfolio already returns 11.06%:  
asking for “at least 8%” costs nothing and the multiplier is zero
- ▶ Only beyond 11.06% does the constraint bite: every additional point of return is paid for in variance (multiplier  $\approx 0.022$  at  $\bar{r} = 12\%$ )
- ▶ Recognizing inactive constraints avoids “paying” for constraints that are free

# The fragility of the estimates

## Observation 1: The fragility of the estimates

*In the simulated data the asset TEC has a true  $\alpha$  of 11% per year, but over 60 months the estimated return is 1.6%: the noise dominates.*

*This is the most important practical lesson of Markowitz: **the estimates of the expected returns are far more unstable than those of the covariances**, and optimized portfolios chase the estimation errors.*

# Remedies used in practice

- ▶ The constraints  $u_i$  on the shares per asset
- ▶ **Shrinkage** of the estimates
- ▶ Optimizing risk only (**minimum variance**)

# Summary

- ▶ Below 11.06% the return constraint is inactive: zero multiplier
- ▶ Beyond it, every point of return costs variance ( $\approx 0.022$  at  $\bar{r} = 12\%$ )
- ▶ The estimates of the expected returns are fragile: optimization chases the estimation errors
- ▶ Remedies: caps, shrinkage, minimum variance

# Exercises

# Introduction: Exercises

- ▶ Five exercises, by hand and with the script of the case study
- ▶ They concern the correlation, the caps per asset, the stability of the estimates, the tracking error and the transaction costs



## Exercise: check

- ▶ Verify by hand the example with two assets and  $\rho = 0.5$ : the formula becomes

$$x_1 = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

- ▶ Is diversification still worthwhile?

## Exercise: frontier

- ▶ Recompute the frontier with  $u_i = 20\%$
- ▶ How much maximum return is lost?
- ▶ Draw the three frontiers ( $u = 1, 0.3, 0.2$ ) in the same plot

## Exercise: stability of the estimates

- ▶ Run the script again with a different random seed (`default_rng(1)`)
- ▶ Compare the optimal compositions for the same  $\bar{r}$ : how much do they change?
- ▶ Which operational implication do you draw from this?

## Exercise: tracking error

- ▶ The benchmark is the equally weighted portfolio  $x_i^B = 1/8$
- ▶ Minimize the tracking error

$$\sum_{i=1}^n \sum_{j=1}^n q_{ij} (x_i - x_i^B)(x_j - x_j^B)$$

with expected return  $\geq 12\%$

- ▶ How far must one depart from the benchmark?

## Exercise: transaction costs

- ▶ Starting from the equally weighted portfolio, add the quadratic cost

$$\kappa \sum_{i=1}^n (x_i - 1/8)^2, \quad \kappa = 0.5$$

- ▶ How does the composition change with respect to the “clean” optimum?

# Summary

- ▶ Effect of the correlation on diversification
- ▶ Cost of the caps per asset on the frontier
- ▶ Stability of the compositions with respect to the data
- ▶ Variants of the model: tracking error with respect to a benchmark and quadratic transaction costs

# Final summary (1/2)

## Model:

- ▶ Convex QP: minimum variance with a minimum return, budget and bounds per asset
- ▶ Equivalent formulations: mean-variance, bounded risk

## Diversification:

- ▶ Combining weakly correlated assets lowers the risk below that of the individual assets

# Final summary (2/2)

## Efficient frontier:

- ▶ Menu of the risk-return trade-offs; the equally weighted portfolio is dominated
- ▶ The mandate constraints have a visible cost

## Estimates:

- ▶ Inactive constraints cost nothing
- ▶ The estimates of the expected returns are fragile: use caps, shrinkage or minimum variance